DBMS LAB 7

Parth Janardan Naukudkar_ECS_II_66_D6

Aim: Understanding and applying INNER JOIN, LEFT OUTER JOIN, RIGHT OUTER JOIN, FULL OUTER JOIN, SELF JOIN, CROSS JOIN, and Multi-table Joins on relational schemas.

1 [Subquery - IN]: Write an SQL query using an IN subquery to find the full names and countries of players who scored more than 50 runs (NRuns > 50) in any single match.

```
SELECT FName, LName, Country
FROM Player
WHERE PlayerID IN (
   SELECT PID
   FROM Batting
   WHERE NRuns > 50
);
```

FName	LName	Country
MS	Dhoni	India
Sanath	Jayasuriya	Sri Lanka
Sachin	Tendulkar	India

2 [Subquery - NOT IN]: Retrieve all players (PlayerID, FName, LName, Country) who have NEVER bowled in any match using a NOT IN subquery referencing the Bowling table.

```
SELECT PlayerID, FName, LName, Country
FROM Player
WHERE PlayerID NOT IN (
   SELECT PID
   FROM Bowling
);
```

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PlayerID	FName	LName	Country
21001	Virender	Sehwag	India
25001	MS	Dhoni	India
27001	Praveen	Kumar	India
27002	Ishant	Sharma	India
89001	Sachin	Tendulkar	India
90001	Brain	Lara	West Indies
90002	Anil	Kumble	India
92001	Shane	Warne	Australia
92003	Inzamam	UI-Huq	Pakistan
93002	Heath	Streak	Zimbabwe
93003	Gary	Kirsten	South Africa
94004	Stephen	Fleming	New Zealand
95001	Ricky	Ponting	Australia
95002	Shaun	Pollock	South Africa
95003	Jacques	Kallis	South Africa
96001	Rahul	Dravid	India
96002	Herschelle	Gibbs	South Africa
96003	Saurav	Ganguly	India
97004	Daniel	Vettori	New Zealand
98001	Shahid	Afridi	Pakistan
99002	Adam	Gilchrist	Australia
99003	Michale	Vaughan	England
300311	Michael	Clark	NULL
406317	Sam	NULL	India
NULL	NULL	NULL	NULL

3 [Subquery - EXISTS]: Write an SQL query using EXISTS to list all matches (MatchID, Ground, Date) where at least one player hit 2 or more sixes (Six >= 2).

```
SELECT M.MatchID, M.Ground, M.Date
FROM ODIMatch M
WHERE EXISTS (
   SELECT *
   FROM Batting B
   WHERE B.MatchID = M.MatchID
    AND B.Six >= 2
```

);

	MatchID	Ground	Date
▶	2755	Colombo	2008-08-24
	2689	Brisbane	2008-03-04
*	NULL	NULL	NULL

4 [Subquery - NOT EXISTS]: Retrieve all players (PlayerID, FName, LName) for whom NO bowling record exists in the Bowling table using a correlated NOT EXISTS clause.

```
SELECT P.PlayerID, P.FName, P.LName
FROM Player P
WHERE NOT EXISTS (
   SELECT *
   FROM Bowling B
   WHERE B.PID = P.PlayerID
);
```

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PlayerID	FName	LName
21001	Virender	Sehwag
25001	MS	Dhoni
27001	Praveen	Kumar
27002	Ishant	Sharma
89001	Sachin	Tendulkar
90001	Brain	Lara
90002	Anil	Kumble
92001	Shane	Warne
92003	Inzamam	UI-Huq
93002	Heath	Streak
93003	Gary	Kirsten
94004	Stephen	Fleming
95001	Ricky	Ponting
95002	Shaun	Pollock
95003	Jacques	Kallis
96001	Rahul	Dravid
96002	Herschelle	Gibbs
96003	Saurav	Ganguly
97004	Daniel	Vettori
98001	Shahid	Afridi
99002	Adam	Gilchrist
99003	Michale	Vaughan
300311	Michael	Clark
406317	Sam	NULL
NULL	NULL	NULL

5 [Scalar Subquery in WHERE]: Find all players (FName, LName, Yborn) who were born before the average birth year (Yborn < AVG(Yborn)) of all registered players.

```
SELECT FName, LName, Yborn
FROM Player
WHERE Yborn < (
   SELECT AVG(Yborn)
   FROM Player
);
```

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	FName	LName	Yborn
▶	Sachin	Tendulkar	1973
	Brain	Lara	1969
	Anil	Kumble	1970
	Sanath	Jayasuriya	1969
	Shane	Warne	1969
	Muthiah	Murlitharan	1972
	Inzamam	UI-Huq	1970
	Heath	Streak	1974
	Gary	Kirsten	1967
	Chaminda	Vaas	1974
	Stephen	Fleming	1973
	Ricky	Ponting	1974
	Shaun	Pollock	1973
	Rahul	Dravid	1973
	Herschelle	Gibbs	1974
	Saurav	Ganguly	1972
	Adam	Gilchrist	1971
	Michale	Vaughan	1974

6 [Subquery - ALL Operator]: Write a query to find player batting performances where NRuns is greater than ALL runs scored by players representing 'Australia'.

```
SELECT B.*
FROM Batting B
WHERE B.NRuns > ALL (
   SELECT B2.NRuns
   FROM Batting B2
   JOIN Player P ON B2.PID = P.PlayerID
   WHERE P.Country = 'Australia'
);
```

	MatchID	PID	POrder	HOut	FOW	NRuns	Mts	Nballs	Fours	Six
▶	2755	25001	5	C	232	71	104	80	4	0
	2755	91001	1	C	74	60	85	52	8	2
	2689	89001	2	C	205	91	176	121	7	0

7 [Subquery - ANY / SOME]: Find all bowling performances where Nwickets is greater than ANY wicket count achieved by bowlers from 'Sri Lanka'.

```
SELECT B.*
FROM Bowling B
WHERE B.NWickets > ANY (
    SELECT B2.NWickets
    FROM Bowling B2
    JOIN Player P ON B2.PID = P.PlayerID
    WHERE P.Country = 'Sri Lanka'
);
```

MatchID	PID	NOvers	Maidens	Nruns	Nwickets
2689	99001	10	0	58	1
2689	24001	3	0	27	1
2755	94002	9	1	40	1
2755	92002	10	0	56	1
2755	23001	10	0	53	2
2755	98002	10	0	40	3
2689	98002	10	0	44	1

8 [Correlated Subquery]: Write a correlated subquery to list each player who scored the maximum runs (NRuns) among all players from their own Country.

```
SELECT P.PlayerID, P.FName, P.LName, P.Country, B.NRuns
FROM Player P
JOIN Batting B ON P.PlayerID = B.PID
WHERE B.NRuns = (
    SELECT MAX(B2.NRuns)
    FROM Player P2
    JOIN Batting B2 ON P2.PlayerID = B2.PID
    WHERE P2.Country = P.Country
);
```

PlayerID	FName	LName	Country	NRuns
91001	Sanath	Jayasuriya	Sri Lanka	60
89001	Sachin	Tendulkar	India	91
24001	Andrew	Symonds	Australia	42

9 [Nested Subquery with HAVING]: Display players whose average runs per match is strictly greater than the overall average runs scored across all matches in the Batting table.

```
SELECT P.PlayerID, P.FName, P.LName,
    AVG(B.NRuns) AS Average_Runs
FROM Player P
JOIN Batting B ON P.PlayerID = B.PID
GROUP BY P.PlayerID, P.FName, P.LName
HAVING AVG(B.NRuns) > (
  SELECT AVG(NRuns)
  FROM Batting
);
```

PlayerID	FName	LName	Average_Runs
25001	MS	Dhoni	53.5000
91001	Sanath	Jayasuriya	60.0000
89001	Sachin	Tendulkar	91.0000
24001	Andrew	Symonds	42.0000

10 [Derived Table in FROM]: Create a derived table in the FROM clause calculating total runs and total balls faced per player, then select players with a strike rate ((Total_Runs / Total_Balls) * 100) greater than 100.

```
SELECT P.PlayerID, P.FName, P.LName,
    T.Total_Runs, T.Total_Balls,
    (T.Total_Runs * 100.0 / T.Total_Balls) AS Strike_Rate
FROM Player P
```

```
JOIN (
   SELECT PID,
        SUM(NRuns) AS Total_Runs,
        SUM(Nballs) AS Total_Balls
   FROM Batting
   GROUP BY PID
) AS T ON P.PlayerID = T.PID
WHERE T.Total_Balls > 0
  AND (T.Total_Runs * 100.0 / T.Total_Balls) > 100;
```

	PlayerID	FName	LName	Total_Runs	Total_Balls	Strike_Rate
▶	91001	Sanath	Jayasuriya	60	52	115.38462

11 [Set Operator - UNION]: Write a query using UNION to combine the PlayerIDs of players who scored > 50 runs in Batting with players who took >= 3 wickets in Bowling.

```
SELECT PID AS PlayerID
FROM Batting
WHERE NRuns > 50
UNION
SELECT PID AS PlayerID
FROM Bowling
WHERE NWickets >= 3;
```

	PlayerID
▶	25001
	91001
	89001
	98002

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12 [Set Operator / Subquery Intersect]: Find players who both batted AND bowled in MatchID = 2755 using an INTERSECT query or equivalent subquery structure.

```
SELECT PID AS PlayerID
FROM Batting
WHERE NRuns > 50
UNION
SELECT PID AS PlayerID
FROM Bowling
WHERE NWickets >= 3;
```

PlayerID
23001
91001
94002
92002
98002

13 [Complex Grouping & HAVING]: Display Ground, total runs scored on that ground, and total matches played, considering only grounds where more than 2 matches took place and total runs exceed 100.

```
SELECT M.Ground,
    SUM(B.NRuns) AS Total_Runs,
    COUNT(DISTINCT M.MatchID) AS Total_Matches
FROM ODIMatch M
JOIN Batting B ON M.MatchID = B.MatchID GROUP BY M.Ground HAVING
COUNT(DISTINCT M.MatchID) > 2 AND SUM(B.NRuns) > 100;
```

Ground	Total_Runs	Total_Matches